Representative Open Datasets

EXSU 500: Fundamentals of Artificial Intelligence in Medicine

Fall 2026 · Team Project · Non-exhaustive

How to use this list

This list is a starting point, not a menu and not a limit. Any dataset that meets the access rules below is acceptable, whether or not it appears here. Equally, appearing here does not mean a dataset suits your question: several entries are large, awkward, or only loosely labelled, and one of the things you are being assessed on is choosing sensibly.

Before you settle on a dataset, download it and load it. A dataset you cannot open in week six is a project you cannot finish in week thirteen. Check the file size against the hardware you actually have, and check that the labels you need are really present rather than merely described in a paper.

Each entry is marked with an access tier:

- Open. Downloadable by anyone from the URL, with no account, no application and no agreement. These are the safest choices for this course.
- Registration. A free account, a click-through licence, or a short form. Usually granted immediately or within a day or two. Acceptable, but start early.
- Restricted. Credentialed access, ethics training, a data use agreement, or an institutional application. These are listed so that you recognise and avoid them. They are not recommended for this course, because their approval cycles are longer than the term.

URLs were checked in August 2026. Dataset hosting moves; if a link is dead, search the dataset name and tell the teaching assistants so the list can be corrected.

1. Starter collections and search portals

If you are unsure where to begin, begin here. The first entry in particular is built for exactly this kind of course project.

Dataset	What it is	Access	URL
MedMNIST v2 / MedMNIST+	Eighteen standardised biomedical image datasets, 2D and 3D, pre-processed to uniform small sizes with labels and official train, validation and test splits. Covers pathology, chest radiography, dermatoscopy, OCT, ultrasound, blood cells, abdominal CT, nodules and more. Trains on a free cloud notebook in minutes. The recommended default for this course.	Open	medmnist.com
PhysioNet	Large repository of physiological signals and clinical data. Many databases are fully open; a subset is credentialed. Filter for open access before you plan around anything here.	Open, mixed	physionet.org/about/database
Grand Challenge	Index of biomedical imaging challenges, many with public training data, defined tasks and published leaderboards. Useful for finding a problem that already has a sensible evaluation protocol.	*Mixed*	grand-challenge.org/challenges
Kaggle Datasets	Very large, uneven collection including many medical datasets and competition sets. Quality and provenance vary widely, so check the original source before trusting the labels.	*Registration*	kaggle.com/datasets
Hugging Face Datasets	Hosted datasets with a uniform download API, including a	Open,	huggingface.co/datase

	growing set of biomedical image, signal and text collections.	mixed	ts
UCI Machine Learning Repository	Classic tabular benchmark repository. Small, clean, well documented clinical datasets that are ideal for a first modelling pipeline.	Open	archive.ics.uci.edu

2. Radiography and cross-sectional imaging

Dataset	What it is	Access	URL
NIH ChestX-ray14	Around 112,000 frontal chest radiographs from about 30,000 patients, labelled for fourteen findings by text mining of reports. Large, so consider a downsampled version. Label noise is a known issue and a good thing to discuss in your limitations.	Open	nihcc.app.box.com/v/ChestXray-NIHCC
CheXpert	About 224,000 chest radiographs from Stanford with uncertainty-aware labels for fourteen observations. Widely used benchmark with an established evaluation protocol.	*Registration*	stanfordaimi.azurewebsites.net
Medical Segmentation Decathlon	Ten 3D segmentation tasks across CT and MRI: brain, heart, liver, hippocampus, prostate, lung, pancreas, vessel, spleen and colon. Standardised format, well defined tasks.	Open	medicaldecathlon.com
The Cancer Imaging Archive	Hundreds of de-identified imaging collections, many with linked clinical and genomic data. Access varies by collection: most are public, a few are restricted.	*Mixed*	cancerimagingarchive.net
BraTS	Multi-institutional multi-parametric MRI with expert brain tumour segmentations. Long-running challenge with published baselines.	*Registration*	synapse.org/brats
RSNA challenge datasets	Curated radiology sets released for annual challenges, including pneumonia detection, intracranial haemorrhage, pulmonary embolism and cervical spine fracture. Hosted through Kaggle.	*Registration*	kaggle.com/organizations/rsna
MURA	About 40,000 musculoskeletal radiographs of upper extremity studies, labelled normal or abnormal by radiologists.	*Registration*	stanfordmlgroup.github.io/competitions/mura
OASIS brain MRI	Cross-sectional and longitudinal neuroimaging with cognitive and clinical variables, oriented toward ageing and dementia research.	*Registration*	sites.wustl.edu/oasisbrains
OpenNeuro	Open archive of MRI, EEG, MEG, iEEG and PET studies in BIDS format, so datasets share a predictable directory structure.	Open	openneuro.org

3. Physiological signals and time series

Dataset	What it is	Access	URL
PTB-XL	About 21,800 clinical 12-lead ECGs of 10 seconds from roughly 18,900 patients, annotated by up to two cardiologists with 71 statements and rich metadata. Supports binary, multi-class and multi-label formulations. One of the best signal datasets available openly.	Open	physionet.org/content/ptb-xl
MIT-BIH Arrhythmia Database	Forty-eight fully annotated half-hour two-lead ECG recordings. Small, canonical, and well suited to beat classification and anomaly detection.	Open	physionet.org/content/mitdb
PhysioNet/CinC Challenge 2020	Multi-source 12-lead ECG collection assembled for the 2020 challenge, with diagnostic labels across several contributing databases.	Open	physionet.org/content/challenge-2020
Sleep-EDF Database Expanded	Polysomnograms with EEG, EOG and other channels, plus expert hypnograms scoring sleep stages. A clean entry point to multi-channel time-series classification.	Open	physionet.org/content/sleep-edfx

Apnea-ECG Database	ECG recordings annotated minute by minute for apnoea, supporting event detection from a single channel.	Open	physionet.org/content/apnea-ecg
EEG Motor Movement/Imagery	EEG from 109 volunteers performing motor and motor-imagery tasks. Standard brain-computer interface benchmark.	Open	physionet.org/content/eegmmidb
Gait in Neurodegenerative Disease	Force-sensor gait recordings from patients with Parkinson disease, Huntington disease and ALS, with healthy controls.	Open	physionet.org/content/gaitndd
CinC/PhysioNet challenge archive	Twenty-five years of annual challenges: sepsis prediction, heart murmur detection, EEG outcome after cardiac arrest, and more. Each has a defined task and a public training set.	Open, mixed	physionet.org/about/challenge

4. Electronic health records and tabular clinical data

Note that the best known critical-care EHR datasets are credentialed and therefore not usable in this course. Two good alternatives are synthetic records and the open tabular sets below.

Dataset	What it is	Access	URL
Synthea synthetic EHR	Synthetically generated but clinically realistic patient records with full longitudinal histories, in FHIR, CSV and other formats. No privacy constraints at all, so you can publish freely. Excellent for pipeline work, weaker for genuine clinical signal.	Open	synthea.mitre.org/downloads
Diabetes 130-US Hospitals	About 100,000 diabetic inpatient encounters over ten years with demographics, diagnoses, medications and readmission outcome. The standard open readmission-prediction dataset.	Open	archive.ics.uci.edu/dataset/296
Heart Disease (Cleveland and others)	Small classic tabular dataset for coronary disease prediction from clinical and exercise variables. Good for a baseline pipeline in an afternoon.	Open	archive.ics.uci.edu/dataset/45
Breast Cancer Wisconsin (Diagnostic)	Cell-nucleus features from digitised fine-needle aspirates with benign or malignant labels. Very small and very well behaved.	Open	archive.ics.uci.edu/dataset/17
Support2	Prognostic dataset on seriously ill hospitalised adults, with survival times, physiology and clinician estimates. Suitable for survival analysis.	Open	archive.ics.uci.edu/dataset/880
MIMIC-IV and MIMIC-IV-ED	The reference critical-care EHR dataset. Listed here so you recognise it: it requires CITI training and a signed agreement, so it is NOT recommended for this course.	*Restricted*	physionet.org/content/mimiciv
eICU Collaborative Research Database	Multi-centre critical care data. Credentialed, and therefore also NOT recommended for this course.	*Restricted*	physionet.org/content/eicu-crd

5. Genomics, transcriptomics and multi-omics

Dataset	What it is	Access	URL
GDC Data Portal (TCGA, TARGET)	Harmonised cancer genomics across 33 tumour types: RNA-seq expression, DNA methylation, miRNA, copy number and mutations, with clinical and survival annotation. Open tiers are downloadable without application; controlled tiers holding raw sequence are not.	Open tier	portal.gdc.cancer.gov
UCSC Xena	The most practical route into TCGA for a course project: pre-assembled per-cohort matrices of expression, methylation, phenotype and survival, downloadable as flat files.	Open	xenabrowser.net/datapages
cBioPortal	More than 300 curated cancer genomics studies with a clean web interface and a documented API. Good when you want a specific cohort rather than everything.	Open	cbioportal.org
Gene Expression	Vast archive of functional genomics series, including large	Open	ncbi.nlm.nih.gov/geo

Omnibus	disease-specific methylation and expression studies. Quality and documentation vary by submission.		
MLOmics	Cancer multi-omics assembled specifically for machine learning: 8,314 samples, 32 cancer types, four omics types, with baselines. Saves weeks of preprocessing.	Open	doi.org/10.1038/s41597-025-05235-x
DepMap	Genomic, transcriptomic and drug-sensitivity profiles across more than a thousand cancer cell lines. Well suited to drug-response regression.	Open	depmap.org/portal/download
GTEx	Gene expression across many human tissues from non-diseased donors. Summary and expression data are open; individual-level genotypes are controlled.	Open tier	gtexportal.org/home/downloads
Human Protein Atlas	Protein expression across tissues, cell lines and cancers, with pathology images and downloadable summary tables.	Open	proteinatlas.org/about/download
1000 Genomes Project	Reference human genetic variation across worldwide populations. Large, and best approached through a targeted subset.	Open	internationalgenome.org/data

6. Dermatology, endoscopy, ophthalmology and pathology

Dataset	What it is	Access	URL
HAM10000	10,015 dermatoscopic images across seven diagnostic categories, with patient age, sex and lesion site. Strongly imbalanced, which makes it a good vehicle for discussing metrics honestly.	Open	doi.org/10.7910/DVN/DBW86T
ISIC Archive	The largest public dermatoscopic collection, including the annual challenge datasets for lesion classification and segmentation.	Open	challenge.isic-archive.com/data
Kvasir-SEG	1,000 colonoscopy polyp images with pixel-level segmentation masks and bounding boxes. Compact enough to train a segmentation model on modest hardware.	Open	datasets.simula.no/kvasir-seg
HyperKvasir	The largest open gastrointestinal endoscopy collection: labelled and unlabelled images, videos and segmentation masks across 23 findings.	Open	datasets.simula.no/hyper-kvasir
DRIVE and STARE	Retinal fundus images with expert vessel segmentations. Small, classic, and still a fair test of a segmentation pipeline.	Open	drive.grand-challenge.org
APTOS 2019 Blindness Detection	Retinal fundus photographs graded for diabetic retinopathy severity on a five-point scale. Ordinal target, which rewards thinking about the loss function.	*Registration*	kaggle.com/competitions/aptos2019-blindness-detection
PatchCamelyon	327,000 histopathology patches from lymph node sections labelled for metastatic tissue. Binary, large and clean.	Open	github.com/basveeling/pcam
BreakHis	About 7,900 breast tumour histopathology images at four magnifications, benign and malignant, with subtype labels.	Open	web.inf.ufpr.br/vri/databases/breast-cancer-histopathological-database-breakhis

7. Clinical and biomedical text

Text projects are viable, but note that most clinical note corpora are credentialed. Biomedical literature and question-answering sets are the open alternative.

Dataset	What it is	Access	URL
PubMed and PMC Open Access	Millions of biomedical abstracts and full-text open-access articles, with bulk download and an API. Suitable for classification, retrieval and summarisation.	Open	ncbi.nlm.nih.gov/pmc/tools/openftlist

MedQuAD	About 47,000 medical question and answer pairs derived from trusted health websites.	Open	github.com/abachaa/MedQuAD
PubMedQA	Biomedical question answering over research abstracts, with yes, no and maybe labels.	Open	pubmedqa.github.io
MedMCQA	More than 190,000 medical entrance-examination questions across 21 subjects. Useful for evaluating language models rather than training from scratch.	Open	medmcqa.github.io
MIMIC-III and MIMIC-IV Notes	De-identified clinical notes. Credentialed, and therefore NOT recommended for this course.	*Restricted*	physionet.org/content/mimic-iv-note

8. Population health and epidemiology

Dataset	What it is	Access	URL
NHANES	US national survey combining interviews, physical examination, and laboratory results. Rich, well documented, and a natural fit for phenotyping and unsupervised work.	Open	wwwn.cdc.gov/nchs/nhanes
BRFSS	Very large annual US telephone survey of health behaviours and chronic conditions, with hundreds of variables per respondent.	Open	cdc.gov/brfss/annual_data/annual_data.htm
Canadian Community Health Survey	Statistics Canada survey of health status, determinants and utilisation. Public use microdata files are available, some through institutional access.	*Mixed*	statcan.gc.ca
WHO Global Health Observatory	Country-level indicators across mortality, disease burden, risk factors and health systems, with an API. Good for global comparisons rather than patient-level prediction.	Open	who.int/data/gho
SEER	US cancer incidence and survival with long follow-up. Requires a signed data-use agreement, so treat it as restricted for this term unless approved in advance.	*Restricted*	seer.cancer.gov/data

9. Before you commit

- Read the license. Some datasets are non-commercial, some require specific citation wording, and some forbid redistribution. Your repository licence must be compatible with your dataset licence.
- Cite the dataset as its custodians ask. Most of the entries above have a paper that must be cited alongside the URL.
- Check for grouping structure. If a dataset has several images or recordings per patient, your splits must be made at the patient level, not the row level.
- Check the class balance before choosing metrics. Several of these datasets are severely imbalanced, which makes accuracy meaningless.
- **Do NOT commit the data itself to your repository**. Provide a download script or documented instructions instead.